MIL-C-60678B(MU)
5 May 1970
SUPERSEDING
MIL-C-60678A(MU)
22 September 1967

MILITARY SPECIFICATION

CARTRIDGE, 20MM, HIGH EXPLOSIVE INCENDIARY TRACE-SELF DESTRUCT, M246

1. SCOPE

1.1 This specification covers one type of electric primed cartridge for use in Gun, 20mm, Automatic, XM168.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-644	- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition
MIL-STD-651	- Visual Inspection Standards for 20mm Ammunition and Components
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

D7258928	- Marking on Shipping and Storage Container, M548, for 20mm Cartridges
F7258941	- Method of Packing 100 Cartridges, 20mm, with M103 Type Case, in Bulk, in M548 Container
F7258942	- Method of Packing 200 Cartridges, 20mm, with M103 Type Case, in Bulk, in M548 Container
F7258969	- Method of Packing 100 Cartridges, 20mm, in M14 or M14A1 Links, in M548 Container

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

2.1 DRAWINGS (contd)

D10542384 - Cartridge, 20mm, HEIT-SD, M246
D10542424 - Projectile, 20mm, HEIT-SD, M246
C10542427 - Projectile, 20mm, HEIT-SD, M246, Charged Body
B10542429 - Seal
IEL 10542384 - Index of Inspection Equipment Lists for: Cartridge, 20mm, HEIT-SD, M246

PUBLICATIONS

U.S. ARMY MATERIEL COMMAND

AMCR 715-505 - Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20mm Cartridges

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-The cartridge shall comply with Drawing (Dwg.) D10542384, referenced specifications, and other requirements specified herein.

3.2 Manufacturing process.-All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without his prior approval. (see 6.3)

3.3 Projectile extraction.-The projectile shall separate from the cartridge case within tension limits specified on Dwg. D10542384.

3.4 Electrical resistance, primed case.-The electrical resistance of the cartridge primer shall be not less than 500 ohms, or more than 1,200,000 ohms.

3.5 Projectile torque.-The projectile shall withstand a minimum torque as specified on Dwg. D10542384.

3.6 Velocity.-The average velocity of the sample cartridges conditioned at 68 degrees Fahrenheit (°F) to 72°F shall be 3380 ± 50 feet per second (fps) at 78 feet from the gun muzzle, and the sample standard deviation shall not exceed 40 fps.

3.7 Pressure.-The average chamber pressure ($\bar{X}$) of the sample cartridge, conditioned at 68°F to 72°F plus 0.84 times the average range ($\bar{R}$) of the chamber pressure shall not exceed 60,500 pounds per square inch (psi).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

3.8 Waterproofness.-The average velocity (wet) of projectiles of the sample cartridges conditioned at 68°F to 72°F shall not vary from the average velocity (dry) by more than 100 fps.

3.9 Accuracy.-The average mean radius of all targets resulting from sample cartridges fired at ambient temperature shall not exceed 15 inches at a range of 600 yards.

3.10 Action time.-The action time of the cartridge shall not exceed 4.0 milliseconds at either ambient or -65°F temperature.

3.11 Projectile trace.-The projectile shall exhibit a trace visible against a dark background for a minimum time of 3.1 seconds when conditioned at -65°F and 145°F unless self destruction occurs earlier.

3.12 Projectile function.

3.12.1 Impact function.-The projectile shall function with high order detonation upon target impact.

3.12.2 Self-destruct function.-The self destruction of the projectile, when conditioned at -65°F and 145°F, shall occur between 3.1 and 7 seconds, inclusive, of uninterrupted flight.

3.13 Function and casualty.-The cartridge, at ambient temperature, shall function without casualty.

3.14 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.14.1 Process defects.-The cartridge shall be free of defects such as folds, wrinkles, buckles, deep draw scratches, cuts, dents, perforations and other forms of substandard workmanship.

3.14.2 Cleanliness.-The cartridge and its components shall be free of all foreign matter. (see 6.2)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 650 cartridges. (see 6.6) The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. Identification shall be in accordance with MIL-STD-1168.

4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample will be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract. (see 6.6)

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications will result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.3.1.1 Lot.-A lot shall consist of:

a. Cartridges loaded by one manufacturer, in one unchanged process, in accordance with the same drawings and drawing revisions, and same specification and specification revision.

b. Like parts and assemblies (cases, primers, projectile metal parts, fuze assemblies), having one manufacturer's symbol and one interfix number.

c. Propellant from one lot.

4.3.1.2 Lot identification.-Each cartridge and each packed ammunition lot shall be identified in accordance with applicable drawings and MIL-STD-1168, supplemented as directed by the procuring agency. Each packed lot shall be further identified by a federal stock number assigned by the procuring activity.

4.3.2 Examination.-One hundred percent examination shall be performed by the contractor for all critical defects. Examination for major and minor defects shall be performed as specified herein. Any cartridge found to be defective shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.2.1 Primed cartridge cases.-Each primed cartridge case shall be tested by the contractor for electrical resistance subsequent to primer crimping and waterproofing, using the method of 4.4.2.1. Primed cases which fail to meet the requirements of 3.4 shall be removed from the lot.

4.3.2.2 Sampling and acceptance criteria.-Except as otherwise indicated, sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235. Defect characteristics may be grouped into a class and considered collectively, or defect characteristics may be considered individually to determine acceptability. When acceptability is determined on a class basis, in addition to the AQL assigned to the major class, an AQL shall be assigned to individual characteristics within the major class, limiting the number of defectives containing a particular major defect to less than the number of defectives permitted for the class. However, where sampling procedures of MIL-STD-1235 are used, acceptability shall not be determined on a class basis for major defects. In addition, where three or less defects appear in a classification, sampling plans applicable to a class shall not be used for major defects. (see 6.5)

4.3.2.3 AQL's.-The following AQL's are assigned to major and minor defects:

a. Individual:

Major ---------- 0.25%
Minor ---------- 0.40%

b. Class basis:

Major ---------- 1.50%
Minor ---------- 2.50%

c. Individual defects associated with the class basis:

Major ---------- 0.40%

4.3.3 Classification of defects 1/.-The classification of defects shall be as follows:

4.3.3.1 Seal (see Dwg. B10542429).

Categories and Defects	Method of Inspection
Critical: None defined.	
Major:	
101. Thickness of seal	Gage
102. Corrosion or foreign matter	Visual
Minor: None defined.	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.3.2 Projectile, charged body prior to assembly of seal (see Dwg. C10542427).

Categories and Defects	Method of Inspection
Critical:	
1. Depth, from metering disc seat to tracer charge, max.	Gage
Major:	
101. Depth, from metering disc seat to tracer charge, min.	Gage
Minor: None defined.	

4.3.3.2.1 Projectile, charged body (see Dwg. C10542427).

Categories and Defects	Method of Inspection
Critical:	
1. Seal missing or perforated	Visual
Major:	
101. Metering disc missing	Visual
102. Metering disc crimp defective	Visual/Manual 2/
Minor: None defined.	

4.3.3.3 Cartridge assembly (see Dwgs. D10542384 and D10542424, a detail of Dwg. D10542384).

Categories and Defects	Method of Inspection
Critical:	
1. Crack or split in projectile	Visual
2. Split or perforated case	Visual
3. Low propellant charge	Gage 3/
4. Crimp missing (case-projectile)	Visual
Major:	
101. Loose primer	Manual
102. Depth of primer seating	Gage
103. Overall length, max.	Gage
104. Improper or incomplete crimp (case-projectile)	Visual
105. Profile and alignment, max.	Gage 4/
106. Corrosion	Visual
107. Head configuration	Visual
108. Metal sliver(s) across primer insulator	Visual
109. Cracked, split, loose or dented fuze nose cap	Visual
110. Length, case head to centerline of crimp	Gage
111. Fuze loose	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.3.3 (contd)

Categories and Defects	Method of Inspection
Minor:	
201. Foreign matter (except corrosion)	Visual (see 6.2)
202. Marking incorrect, illegible, or missing	Visual
203. Gap between fuze and projectile body, max.	Gage
204. Protective coating damaged or incomplete	Visual
205. Waterproofing missing around primer	Visual
206. Scratch, dent, bulge, wrinkle or fold in case	Visual
207. Lacquer on primer button	Visual
208. Rotating band damaged	Visual
209. Primer crimp missing or incomplete	Visual
210. Incorrect type cartridge	Visual

1/ MIL-STD-651 shall be used, when applicable, in defining and evaluating cartridge visual defects.

2/ If the crimp closure is not pressed tightly against the metering disc for 360° or the metering disc can be moved manually, the charged body subassembly shall be classed defective.

3/ A cartridge containing 550 grains or less of propellant shall be classed defective.

4/ A dead weight load of 60 pounds maximum may be used to insert the cartridge into the gage.

4.3.4 Tests.-The tests listed in Table I shall be performed on each cartridge lot in accordance with the test methods prescribed in 4.4. Sample size and acceptance criteria for each test shall be as specified. Only cartridges having met the visual and dimensional requirements and having been selected in such a manner that the sample is representative of the entire lot shall be used in the tests. The selected cartridges shall be thoroughly mixed before being divided into samples for the various tests. (see 6.7)

Table I

Test 1/		Sample Size Production Lot		Rqmt Para
Projectile extraction	2/	20		3.3
Electrical resistance verification	3/	20	14/	3.4
Propellant contamination and charge weight verification	4/	20	14/	3.15.2
Projectile torque	5/	20		3.5
Velocity	6/	20		3.6
Pressure	7/	20		3.7

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.4 Table I (contd)

Test 1/	Sample Size Production Lot		Rqmt Para
Waterproofness	8/ 20		3.8
Accuracy	9/ 50		3.9
Action time	10/		3.10
Ambient temperature	25		
-65°F	25		
Projectile trace	12/	15/	3.11
-65°F temperature	110		
+145°F temperature	110		
Projectile function			
Impact function	11/ 50		3.12.1
Self-destruct	12/	15/	3.12.2
-65°F temperature	110		
+145°F temperature	110		
Function and casualty	13/ 200		3.13

1/ The lot shall be rejected if in any firing test one or more of the following malfunctions or casualties of Table II occur:

a. Misfire associated with e or f of Table II, Note 2/.
b. Projectile remaining in bore.
c. Metal parts separation, except rotating band separation (see Note 6/, Table II).

Except as otherwise specified and in the function and casualty test where the occurrence of a firing defect listed in Table II prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired.

2/ Projectile extraction.

a. Failure of two or more sample cartridges to comply with the minimum requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply with the minimum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined samples fail to comply with the minimum requirement.

b. Failure of four or more cartridges to comply with the maximum requirement shall cause rejection of the lot. If more than one but less than four cartridges fail to comply with the maximum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if five or more cartridges of the combined samples fail to comply with the maximum requirement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.4 (contd)

3/ Electrical resistance verification.-The lot will be rejected if any primer of the sample cartridges fail to comply with either of the primer resistance requirements.

4/ Propellant contamination and charge weight verification.-The presence of foreign matter in the propellant of any of the test cartridges or a propellant charge weighing less than 550 grains will cause rejection of the lot. (see 3.14.2, 6.2 and 4.4.3)

5/ Projectile torque.-Failure of two or more of the sample cartridges to comply with the specified torque requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined samples fail to comply with the applicable requirement.

6/ Velocity.-If the sample fails to comply with either or both requirements a retest sample of 40 cartridges shall be tested for the failing parameter(s) (average, sample standard deviation). The lot shall be rejected if the retest sample fails to comply with the applicable requirement(s). (see 6.4.1)

7/ Pressure.-If the average pressure ($\bar{X}$) plus 0.84 times the average range ($\bar{R}$) of the sample cartridges exceeds 60,500 psi, a retest sample of 40 cartridges shall be tested. The lot shall be rejected if the average pressure ($\bar{X}$) plus 0.91 times the average range ($\bar{R}$) of the retest sample fails to comply with the applicable requirement. (see 6.4.2)

8/ Waterproofness.-If the average velocity (wet) of the sample fails to comply with the applicable requirement, a retest sample of 40 cartridges shall be tested. The lot shall be rejected if the average velocity (wet) differs by more than 100 fps from the average velocity (dry) of 40 additional sample cartridges fired to establish a new base for comparison. The lot shall not be penalized for failure of the last named sample cartridges to comply with the requirements of 3.6.

9/ Accuracy.-If the average mean radius of the sample cartridges exceeds the applicable requirement, a retest sample of 100 cartridges shall be tested. The lot shall be rejected if the average mean radius of 10 retest accuracy targets exceeds the applicable requirements.

10/ Action time.-For acceptance, each cartridge of the test sample shall exhibit an action time not exceeding 4 milliseconds (0.004 second) and the sample average action time ($\bar{X}$) plus four sample standard deviations (S) shall not exceed 4 milliseconds. These criteria apply separately to both the ambient and cold phase tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.4 (contd)

11/ Impact function.-Failure of five or more sample cartridges to comply with the applicable requirement shall cause rejection of the lot. If more than two but less than five fail to comply, a second sample of 50 cartridges shall be tested. The lot shall be rejected if seven or more cartridges of the combined samples fail to comply with the applicable requirement.

12/ Projectile self-destruct and trace.-For acceptance, each temperature phase shall be considered individually. Failure of 10 or more sample cartridges to comply with the requirement shall cause rejection of the lot. If more than six and less than 10 cartridges of the sample fail to comply with the requirement, a second sample of 110 cartridges shall be tested for that temperature phase. The lot shall be rejected if 16 or more cartridges of the combined sample fail to comply with either test requirement.

13/ Function and casualty.-The lot shall be rejected if any malfunction or firing casualty of Table II occurs in number(s) equal to or greater than the applicable "Rej" number. Except as otherwise provided, if malfunctions or casualties occur in excess of the applicable "Acc" number, but less than the applicable "Rej" number, a second sample of 400 cartridges shall be fired. The lot shall be rejected if in the combined samples, malfunctions or casualties occur in numbers equal to or greater than the applicable "Rej" number.

14/ Propellant verification.-Primed cases and propellant from the projectile extraction test will be used as samples necessary for the verification tests for electrical resistance, propellant contamination and propellant charge weight. Verification of primer electrical resistance will be performed using the method of 4.4.2.2.

15/ Trace.-The test for trace shall be conducted simultaneously with the self-destruct test.

4.3.4.1 Firing defects.-Firing defects and acceptance criteria shall be as specified in Table II. For defect definitions, see AMCR 715-505, Volume 8.

Table II

Firing Casualties	Acc	Rej
Cartridge:		
Fire "out of battery," 1/	0	1
Misfire 2/	-	-
Failure to chamber	0	1
Failure to extract	0	1
Projectile remaining in bore	0	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.4.1 Table II (contd)

Firing Casualties	Acc	Rej
Primer:		
Missing primer, or button missing from primer	0	2
Primer leak	9	25
Case:		
Longitudinal split 3/		
I or S	10	31
J	4	11
K	0	2
L or M	0	1
Circumferential rupture, partial 3/		
S, J or K	4	11
L	0	1
Circumferential rupture, complete 3/	0	1
Detached metal 4/	0	1
Projectile:		
Rotating band separation 5/		
Complete or partial separation having band seat serration imprint	0	3
Separation having length greater than 0.5 inch, and without band seat serration imprint	1	6
Metal parts separation, except rotating band separation 6/	0	1
Premature functioning 7/	0	1

1/ A cartridge shall be classed defective if it fires, due to delayed ignition, after the bolt has been unlocked and extraction has been initiated or completed.

2/ Misfiring test cartridges, excepting those from the waterproofness test (Table I), shall be retested in accordance with the method of 4.4.11. If a cartridge fails to function in this test, examinations shall be conducted for the following conditions:

a. Excessive primer seating depth.
b. Metal slivers across primer insulator.
c. Electrical continuity.
d. Missing or obstructed case vent.
e. Missing or insufficient primer composition or propellant.
f. Partially burned or contaminated primer composition or propellant.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.3.4.1 (contd)

The lot shall not be penalized if condition a is observed, however, the condition shall be reported for information. The "Acc" number shall be zero and the "Rej" number three, for misfires associated with conditions b, c, or d. The second sample for evaluation of misfires associated with conditions b, c, or d shall consist of the function and casualty second sample (see Table I, Note 14). The lot shall be rejected if either condition e or f is observed. Misfire(s) occurring in the waterproofness test shall be handled in accordance with AMCR 715-505, Volume 8.

3/ See Dwg. C7258801 (a detail of IEL 10542384) for classifying splits and ruptures in fired cartridge cases. If a longitudinal split or circumferential rupture (partial) extends into two or more defined areas, only the mose severe defect criterion of Table II for the areas involved, shall apply. If a rupture results in separation of the cartridge case into two or more portions, the defect shall be classed as a complete circumferential rupture.

4/ Metal sheared or missing from the fired cartridge case exterior, such as rim or neck shears, shall be classed as a defect. The lot shall not be penalized for shavings of metal from the interior wall of the case neck in the crimped area.

5/ For classification as a defect, there must be evidence either by recovery of the band, or portion thereof, or by hole(s) in the fragmentation screen(s). The lot shall not be penalized for normal band fringing, or the separation of band fragments having no evidence of band seat serration imprint and having length of 0.5 inch or less.

6/ Separation or breakup of projectile part(s), except rotating band separation, as evidenced by recovery of the part(s) or fragment(s), or hole(s) in the fragmentation screen(s) shall be classed as a defect.

7/ This defect is further defined as occurring during uninterrupted flight at a distance less than 200 yards from the gun muzzle.

4.3.4.2 Unlisted firing defects.-The lot shall be suspended and referred to the contracting officer for disposition, if a malfunction or casualty not covered by this specification occurring in any firing test indicates that the product is unsuited for the purpose intended.

4.3.5 Packaging, packing and marking inspection.-Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644, as applicable.

4.3.6 Inspection equipment.-Index of Inspection Equipment List No. IEL 10542384 identifies the applicable inspection equipment list required to perform examination and tests prescribed herein. Equipment design shall be in accordance with code designation on the applicable IEL (reference is made to inspection equipment list for definition of coding format).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.4 Test methods and procedures.

4.4.1 Projectile extraction.-The method of test shall be as prescribed in AMCR 715-505, Volume 8.

4.4.2 Electrical resistance.

4.4.2.1 Primed cartridge cases.-The equipment and circuitry specified in IEL 10542384 shall be used to measure the resistance of the primer in each primed cartridge case.

4.4.2.2 Verification.-A Simpson Volt-Ohm-Ammeter, Model 260, or approved equal, shall be used to measure the resistance of the primer in each primed cartridge case from the projectile extraction test sample. Only the RX 100 and RX 10,000 scales of the meter shall be used in order to limit the applied voltage and current, respectively, to 7.5 volts and 1.4 milliamperes, maximum.

4.4.3 Propellant contamination and charge weight verification.-The propellant charge from each of the sample cartridges tested for projectile extraction shall be examined visually for contamination as prescribed in AMCR 715-505, Volume 8. The propellant charge from each of the sample cartridges shall be weighed.

4.4.4 Projectile torque.-Equipment shall be as specified in IEL 10542384. Each cartridge of the test sample shall be marked with a light scratch extending axially across the rotating band onto the case neck. Torque specified shall be applied slowly to the projectile. Movement of the projectile with respect to the cartridge case, detectable by misalignment in the scratch mark, shall be recorded.

4.4.5 Velocity, pressure and waterproofness.-The method of test shall be as prescribed in AMCR 715-505, Volume 8. Statistics necessary for testing of results against acceptance criteria shall be computed. (see 6.4)

4.4.6 Accuracy.-The method of test and measurement of targets shall be as prescribed in AMCR 715-505, Volume 8. The specified range may be shortened in decrements of 100 yards, to a minimum range of 200 yards. For ranges less than 600 yards the average mean radius shall be reduced in direct proportion to the reduction in range.

4.4.7 Action time.-The method of test shall be as prescribed in AMCR 715-505, Volume 8. Statistics necessary for testing of results against the acceptance criterion shall be computed. (see 6.4)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

4.4.8 Projectile trace.-The test for trace shall be conducted simultaneously with self-destruct. The sample cartridges for each temperature phase shall be conditioned for a minimum time of 6 hours. The determination for trace time shall be made at the gun position.

4.4.9 Projectile function.

4.4.9.1 Impact function.-The method of test shall be as prescribed in AMCR 715-505, Volume 8. The aluminum target (2024T4, T3 or T36) of .040 inch nominal thickness shall be located 200 ± 25 yards from the gun muzzle, normal to the line of fire. Observation of projectile functioning shall be made from the camera position.

4.4.9.2 Self-destruct function.-The method of test shall be as prescribed in AMCR 715-505, Volume 8. Test for self destruction shall be conducted simultaneously with the trace test. The sample cartridges for each temperature phase shall be conditioned for a minimum time of 6 hours. The determination for self-destruct time shall be made at the gun position.

4.4.10 Function and casualty.-The method of test shall be as prescribed in AMCR 715-505, Volume 8. The test sample shall be fired in bursts of 50 cartridges. The gun barrel(s) shall be at ambient temperature at the beginning of test and cooled to ambient after each 100 rounds. The cyclic rate of each burst shall not be less than 3000 shots per minute in the gun. The cyclic rate for each burst shall be recorded.

4.4.11 Misfires.-Misfiring cartridges shall be retested in test equipment specified by the IEL for the accuracy test. Results of examinations required by Table II shall be recorded.

4.4.12 Defective weapon.-If a test weapon fails to comply with the minimum cyclic rate during a normal burst operation or is found to be defective during or at the end of burst the test shall be suspended. The weapon shall be corrected or replaced and the test repeated in whole or in part, as indicated. Any firing defect that occurred shall be valid for lot penalty in accordance with Table II unless the defect is found to have been caused by the defective weapon.

5. PREPARATION FOR DELIVERY

5.1 Packing, Level A.-The type of ammunition pack shall be in accordance with instructions from the procuring agency. Linked ammunition shall be packed in accordance with Dwg. F7258969. Bulk ammunition shall be packed in accordance with Dwg. F7258941 or F7258942.

5.2 Marking.-Marking on the shipping and storage container shall be in accordance with Dwg. D7258928.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Ordering data.-Invitations for bids and contracts or orders will specify the following:

6.1.1 Title, number, and date of this specification.

6.1.2 Place of inspection, if not at place of manufacture.

6.1.3 First article sample. (see 4.3 and 6.7)

6.1.4 Detailed packing and marking instructions. (see 5.1 and 5.2)

6.1.5 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.6 Provision for the submission and approval of the manufacturing process. (see 3.2)

6.2 Contamination of explosive components.-Particular care should be exercised to avoid contamination of primers and propellant by oil, grease, or other foreign matter since such contamination can cause misfire(s) or increase action time and lot rejection.

6.3 Process deviation.-A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which may alter the metallurgical or physical properties of the item.

6.4 Computations.

6.4.1 Standard deviation.-Where computation of a sample standard deviation is specified for determination of lot acceptance, the method of computation will be:

$$S = \sqrt{\frac{\Sigma(X_i - \overline{X})^2}{n}} \quad \text{or equivalent}$$

6.4.2 Range plan.-Where a range plan is specified, the individual values will be recorded in subgroups of five in the order in which they are observed. The ranges for the subgroups will be added and the sum of the ranges will be divided by the number of subgroups to determine the average range ($\overline{R}$).

6.5 AQL's.-The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitation, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

6.6 First article sample.-In order to minimize inspection costs, the firing tests of 4.4 will be performed after the sample has been provisionally accepted by the Government for all other requirements. Additional cartridges may be required by the test facility.

6.7 Combining tests.-With approval of the procuring agency, tests may be performed concurrently on the same sample cartridges provided that the test results are not affected by this procedure. The test facility should make maximum use of this procedure to minimize testing costs.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1305-A603

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:21Z
Check the source to verify that this is the current version before use.